

Responsive Web Design

ResNet Academy | Frontend Development

Essential responsive setup

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

```
img, video { max-width: 100%; height: auto; }
*, *::before, *::after { box-sizing: border-box; }
```

Mobile-first media queries

```
/* base = small screens */
.cards { display: grid; gap: 1rem; }

@media (min-width: 40rem) { .cards { grid-template-columns: repeat(2, 1fr); } }
@media (min-width: 64rem) { .cards { grid-template-columns: repeat(3, 1fr); } }
```

- Base styles target the smallest screen. Add **min-width** queries as space grows.
- Pick breakpoints where the **content** breaks. Use **rem**, not device lists.
- Modern range syntax also works: **@media (width >= 48rem)**.

Preference and feature queries

Query	Use
(prefers-color-scheme: dark)	Dark theme palette
(prefers-reduced-motion: reduce)	Disable animation and movement
(hover: hover)	Hover-only effects on devices that can hover
(orientation: landscape)	Landscape-specific tweaks
print	Print stylesheet

Fluid sizing functions

```
width: min(100% - 2rem, 70rem); /* cap the width, keep gutters */
font-size: clamp(1.75rem, 5vw, 3rem); /* min, preferred, max */
grid-template-columns: repeat(auto-fit, minmax(16rem, 1fr)); /* no media query */
aspect-ratio: 16 / 9;
margin-inline: auto;
```

Responsive images

```


<picture>
  <source media="(min-width: 48rem)" srcset="wide.jpg">
  
</picture>
```

Custom properties and theming

```
:root { --color-bg: #fff; --color-text: #0f172a; --space: 1rem; }
@media (prefers-color-scheme: dark) {
  :root { --color-bg: #0b1220; --color-text: #e2e8f0; }
}
body { background: var(--color-bg); color: var(--color-text); padding: var(--space); }
.btn { background: var(--brand, #1d4ed8); } /* fallback after the comma */
```

Motion

```
.btn { transition: background-color .2s ease, transform .2s ease; }
@keyframes fade-in { from { opacity: 0; } to { opacity: 1; } }
@media (prefers-reduced-motion: reduce) { * { animation: none; transition: none; } }
```

Key idea. Animate **transform** and **opacity**. Animating **width**, **height** or **top** forces layout recalculation on every frame.

Testing checklist

- No horizontal scroll at 320px. Check portrait and landscape.
- Touch targets around 44 by 44 CSS pixels or larger, with spacing.
- Test with larger browser text, dark mode and reduced motion.
- Verify on at least one real phone. DevTools device mode is not a substitute.